# John Carlo Z. Cacao

**Mobile Developer · Computer Science Student**

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## PROFESSIONAL SUMMARY

Results-oriented Mobile Developer and Computer Science undergraduate with hands-on experience shipping production-grade applications across Android and iOS, alongside backend service engineering in fintech (PawnHero). Proficient in modern architectures (MVVM, MVI), on-device machine learning with Google ML Kit, native home and lock-screen widgets, and reactive concurrency with Kotlin Coroutines and Dart. Champion and multi-time finalist in collegiate hackathons with a proven ability to solve complex mobile systems challenges.

## WORK EXPERIENCE

**Backend Developer Intern** · PawnHero — June 2026 — Present

- Developing backend services, core financial transaction pipelines, and RESTful API integrations in a high-compliance fintech environment.
- Engineering and optimizing relational database operations with SQL to support operational dashboard tooling and high-concurrency requests.
- Collaborating in cross-functional agile sprints, peer code reviews, and system reliability testing to ensure low-latency performance.

**Service Crew** · McDonald's — April 2025 — May 2026

- Delivered high-volume customer service and operational efficiency during peak shifts, demonstrating exceptional teamwork and adaptability.
- Managed concurrent tasks under time pressure, sharpening rigor, prioritization, and communication while pursuing full-time CS studies.

## FEATURED PROJECTS

**Glot** — On-Device Real-Time Screen Translator — Android

*Kotlin, Jetpack Compose, Google ML Kit (OCR & Translation), MediaProjection API, Coroutines, StateFlow*

- Architected real-time on-device screen translation utility leveraging MediaProjection for low-latency capture and ML Kit for offline multi-script OCR.
- Built custom text tracking using IoU matching and hysteresis-based stabilization, eliminating overlay flicker within tuned pixel deadzones.
- Implemented a "Forbidden Zone" geometry cache preventing OCR feedback loops over active tooltip bounding boxes.
- Engineered a 65% downscaling and bitmap pooling pipeline, cutting memory footprint and latency by over 50%.

**Paddi** — Sticky Notes & Desktop Widget App — Google Play · App Store

*Flutter, Android Home Screen & iOS 18 Lock Screen Widgets, Dart, Local Storage*

- Published production mobile utility providing persistent, interactive desktop and lock-screen sticky note widgets with custom styling.
- Engineered native widget communication channels for instant two-way note state updates and offline-first local persistence.

**Lapi** — Productivity & Task Planning Suite — Google Play · App Store

*Flutter, Elixir, Phoenix Framework, PostgreSQL, MVVM Architecture*

- Designed and deployed cross-platform productivity application with interactive calendar integrations and state-driven task tracking.
- Integrated high-throughput Elixir/Phoenix backend APIs and PostgreSQL persistence for synchronized user workflows.

## TECHNICAL SKILLS

**Languages:** Kotlin, Dart, Elixir, Python, SQL, Java
**Mobile & Frontend:** Android SDK, Jetpack Compose (Material 3), Flutter, Android & iOS Widgets, XML
**Architecture & Concurrency:** MVVM, MVI, Clean Architecture, Kotlin Coroutines, StateFlow, Retrofit, Room, Dagger-Hilt
**Backend & Tools:** Phoenix Framework, PostgreSQL, Firebase, Git/GitHub, Android Studio, Xcode, Postman

## HONORS & AWARDS

**Champion** — CCC DCI Hackathon 2025
**Finalist** — CCC DCI Hackathon 2026
**Finalist** — CICS Hackathon 2026

## EDUCATION & CERTIFICATIONS

**City College of Calamba**
Bachelor of Science in Computer Science (Undergraduate) • Calamba, Laguna

**Computer System Servicing (NC II)**
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